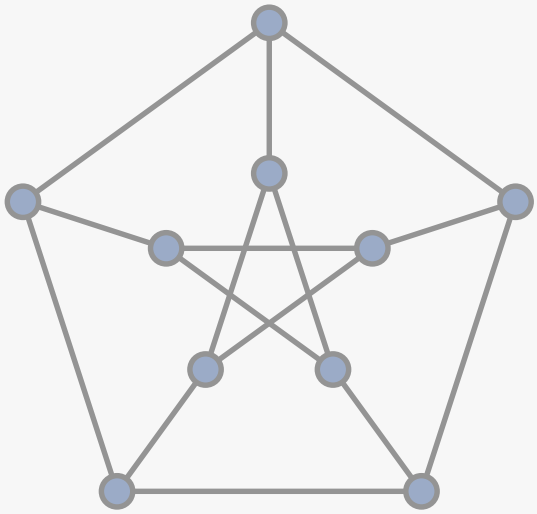
山东大学
SHANDONG UNIVERSITY

The Second Sino-French International Conference on Graph Theory, Combinatorics and Algorithms

Conference Schedule



18–23 October 2026

Jinan, China

数学国家高层次人才培养中心
Mathematical Research Center | Shandong University

山东大学数学学院
School of Mathematics Shandong University

數據科學研究院
Data Science Institute | Shandong University

Conference Schedule

Sunday, 18 October

Venue: 1/F, Xueren Hotel, Central Campus, Shandong University
山东大学中心校区学人大厦一层大堂

Beijing Time	Arrangements
09:00–18:00	**Registration**

Monday, 19 October

Venue: Yifan Hall, 1/F, Building 2, **Hongjialou** Campus, Shandong University
山东大学**洪家楼**校区 2 号楼一层懿范堂

Beijing Time	Speaker	Title	Host
08:30–09:00	**Opening Ceremony**		Guanghui Wang
09:00–09:40	**François Pirot**	The PCF Colouring Conjecture is true for large Δ	Donglei Yang
09:40–10:20	**Hoang La**	Graph Reconstruction: Theory, Efficiency, and Robustness	
10:20–10:40	**Break**		
10:40–11:20	**Michał Seweryn**	Polynomial Bounds for the Graph Minor Structure Theorem	Tong Li
11:20–12:00	**Yangyang Cheng**	Transversal Graph Theory: Progress and Conjectures	
12:00–14:00	**Lunch**		
14:00–14:40	**Paul Bastide**	Fast exploration of temporal graphs	Wanting Sun
14:40–15:20	**Yun Wang**	Disjoint cycle problems in digraphs	
15:20–15:40	**Break**		
15:40–16:20	**Lucas Picasarri-Arrieta**	A generalisation of Ramsey's theorem and its applications to χ- and degree-boundedness	Lin Sun
16:20–17:00	**Ruilong Liu**	Recent Progress on Hypergraph Ramsey Theory	

Conference Schedule

Tuesday, 20 October

Venue: Yifan Hall, 1/F, Building 2, **Hongjialou** Campus, Shandong University
山东大学**洪家楼**校区 2 号楼一层懿范堂

Beijing Time	Speaker	Title	Host
09:00–09:40	**Tao Feng**	On flag-transitive generalized polygons	Linpeng Zhang
09:40–10:20	**Shaoshi Chen**	Flip Graphs for Polynomial Multiplication	
10:20–10:40	**Break**		
10:40–11:20	**Thomas Karam**	A min-max principle for matrix and tensor completion	Jie Hu
11:20–12:00	**Fan Yang**	Small circumference in regular sublinear expanders	
12:00–14:00	**Lunch**		
14:00–17:00	**Problem Session**		Xiangxiang Nie

Wednesday, 21 October

Venue: Zhixin Building / Mingde Building, **Central Campus**, Shandong University
山东大学**中心校区**知新楼/明德楼

Beijing Time	Arrangements
09:00–18:00	**Working Groups**

Conference Schedule

Thursday, 22 October

Venue: Yifan Hall, 1/F, Building 2, **Hongjialou** Campus, Shandong University
山东大学**洪家楼**校区 2 号楼一层懿范堂

Beijing Time	Speaker	Title	Host
09:00–09:40	**Jie Ma**	Longest cycles in highly connected graphs	Zixiang Xu
09:40–10:20	**Bartłomiej Kielak**	Minimal Cayley graphs with large girth and vanishing independence ratio	
10:20–10:40	**Break**		
10:40–12:00	**Working Groups**		
12:00–14:00	**Lunch**		
15:00–15:40	**Hong Liu**	The Erdős-Gallai bound for consecutive even cycle lengths	Yangyang Cheng
16:10–16:50	**Marcin Briański** Online	Onions, Potatoes, and measuring directed graphs without wall immersions	
17:00–18:00	**Candida Bowtell** Online	Andersen's rainbow path conjecture	

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Conference Schedule

Friday, 23 October

Venue: Yifan Hall, 1/F, Building 2, **Hongjialou** Campus, Shandong University
山东大学**洪家楼**校区 2 号楼一层懿范堂

Beijing Time	Speaker	Title	Host
09:00–09:40	**Xizhi Liu**	Extensions of the Andrásfai–Erdős–Sós Theorem	Weihua He
09:40–12:00	**Progress Reports**		Wenling Zhou
12:00–14:00	**Lunch**		
14:00–18:00	**Working Groups (Central Campus)**		

Abstracts

Fast exploration of temporal graphs

Paul Bastide
Technical University of Darmstadt

A temporal graph G is a sequence of graphs $G_1, G_2, \ldots, G_t$ on the same vertex set. In this talk, we are interested in the analogue of the Travelling Salesman Problem for temporal graphs. It is referred to in the literature as the Temporal Exploration Problem, and asks for the minimum length of an exploration of the graph, that is, a sequence of vertices such that at each time step t, one either stays at the same vertex or moves along a single edge of G_t.

One natural and still open case is when each graph Gt is connected and has bounded maximum degree. We present a short proof that any such graph admits an exploration in $O(n^{3/2} \log n^{1/2})$ time steps. In fact, we deduce this result from a more general statement by introducing the notion of average temporal maximum degree. This more general statement improves the previous best bounds, under a unified approach, for several studied exploration problems.

This is based on joint work with Carla Groenland, Lukas Michel and Clé ment Rambaud and Paulina Smolarova.

Biography. Paul Bastide did his PhD with Marthe Bonamy and Carla Groenland on local structures in combinatorics. He did a 1-year postdoc in Oxford with Peter Keevash, and he is now a postdoc with Michal Pilipczuk in Darmstadt.

Andersen's rainbow path conjecture

Candida Bowtell Online
University of Birmingham

In 1989, Andersen conjectured that every properly edge-coloured complete graph on n vertices contains a rainbow path covering at least n-1 vertices. That is, a path which uses any colour at most once. We confirm this conjecture for sufficiently large n, moreover, showing that when at least n colours are used, one can find a rainbow Hamilton path. Our proof combines results of Montgomery on rainbow matchings in bipartite graphs with a novel relaxation of a sorting network. This is joint work with Richard Montgomery, Alp Müyesser and Alexey Pokrovskiy.

Biography. Candida Bowtell has been an Assistant Professor at the University of Birmingham since September 2025. She previously held a Leverhulme Early Career Fellowship and before that completed a postdoctoral position with Professor Richard Montgomery (Warwick) and completed her DPhil at the University of Oxford, supervised by Professor Peter Keevash. Her research interests lie in extremal and probabilistic combinatorics, and many of her projects concern finding large structures in graphs and hypergraphs under certain natural constraints.

Onions, Potatoes, and measuring directed graphs without wall immersions

Marcin Briański Online
Jagiellonian University

Treewidth is among the most important width parameters, largely due to its central role in the Graph Minors project of Robertson and Seymour. Its structural significance comes from the Grid Theorem: the graphs excluding a fixed wall as a minor are exactly those of bounded treewidth. For immersions, the edge-disjoint analogue of minors, Wollan introduced tree-cut width and proved the matching characterisation in terms of exclusion of a wall as an immersion. Extending this theory to directed graphs is substantially more difficult. Nevertheless, Kawarabayashi and Kreutzer's Directed Grid Theorem characterizes bounded directed treewidth by the exclusion of a cylindrical wall as a butterfly minor.

We complete the picture by introducing directed tree-cut width and establishing that the digraphs excluding a fixed cylindrical wall as an immersion are exactly those of bounded directed tree-cut width. Along the way we prove the Potato Theorem, a general tool that decomposes any digraph along a tree into k-flow-connected pieces separated by cuts such that the cycles crossing the cuts have a small edge hitting set.

This talk is based on joint work with Meike Hatzel, Karolina Okrasa, and Michał Pilipczuk.

Biography. Marcin Briański is a mathematician affiliated with Jagiellonian University in Kraków, where he recently completed his PhD thesis under the supervision of Piotr Micek. His research focuses on structural graph and matroid theory, in particular width and depth parameters of graphs, digraphs, and matroids, and the dualities between them.

Flip Graphs for Polynomial Multiplication

Shaoshi Chen
Chinese Academy of Sciences

Flip graphs were recently introduced in order to discover new matrix multiplication methods for matrix sizes. The technique applies to other tensors as well. In this talk, we explore how it performs for polynomial multiplication. This is a joint work with Manuel Kauers.

Biography. Shaoshi Chen is a full professor of the Academy of Mathematics and Systems Science, Chinese Academy of Sciences. His research focuses on symbolic computation, automated theorem proving, and combinatorics. He has published more than 40 papers in ISSAC proceedings and in journals including Foundations of Computational Mathematics, Algebra & Number Theory, and Mathematische Zeitschrift. He has been awarded the Wen-Tsun Wu Award for Young Scholars in Computer Mathematics (2019), the ISSAC Distinguished Paper Award (2021), and the ACA Early Researcher Award (2022).

Transversal Graph Theory: Progress and Conjectures

Yangyang Cheng
University of Passau

Let $G_1, \ldots, G_t$ be graphs on a common vertex set V, and let H be a graph with t edges. A transversal copy of H is a copy of H whose edges can be assigned bijectively to the graphs $G_1, \ldots, G_t$, with each edge belonging to its corresponding graph. Suppose that a graph-theoretic condition P guarantees the existence of H in a single graph. A natural question is whether the same condition, imposed on each of $G_1, \ldots, G_t$, guarantees the existence of a transversal copy of H. In this talk, I will survey recent progress and important open problems in this direction, as well as present some of our contributions to the area. I will also propose several new conjectures that extend and generalize a number of classical conjectures in graph theory.

Biography. Yangyang Cheng, Postdoc at the University of Passau, Germany. PhD from the University of Oxford. He has published several papers in JCTB, JCTA, Forum. Math. Sigma, etc.

On flag-transitive generalized polygons

Tao Feng
Zhejiang University

The classification of flag transitive generalized polygons is listed one of the ten most famous problems in van Maldeghem's monograph Generalized Polygons. In this talk, I summarize the recent progress and open problems in this direction, focusing on generalized triangles (projective planes) and generalized quadrangles.

Biography. Tao Feng is a Professor at the School of Mathematical Sciences, Zhejiang University. He received his Ph.D. from Peking University in 2008 and joined Zhejiang University in 2011, after holding postdoctoral positions in Singapore and the United States. His research interests include algebraic combinatorics and finite geometry. He is a recipient of the National Science Fund for Distinguished Young Scholars, as well as the Kirkman Medal and Hall Medal of the Institute of Combinatorics and its Applications.

A min-max principle for matrix and tensor completion

Thomas Karam
Shanghai Jiao Tong University

It is known that the lowest rank arising from changing the diagonal entries of a matrix and the highest rank arising from restricting the matrix to an off-diagonal submatrix are in a sense two sides of the same coin, as they stand within a multiplicative factor of at most 3. Our focus in this talk shall be twofold.

First, we shall study how the above min-max principle can be extended to more general sets of modifiable matrix entries than the diagonal. If U is a set of entries, then does the highest rank of U-avoiding submatrices control the lowest rank of the matrix after modification of

entries in U? As stated, this is far from being true uniformly over all sets U, but there is an adequate formulation that accounts for the complexity of U, in a sense made precise.

Second, we shall study the extension of this principle to the various rank notions on tensors that can be built from partitioning the coordinates. After recalling a known generalisation to the ranks of tensors of the matrix min-max principle with respect to the diagonal, we will examine increasingly general classes of notions of rank and of sets of modifiable tensor entries, and describe how the formulation of the min-max principle adapts along the path of successive generalisations.

Biography. Prior to becoming an Associate Professor at Shanghai Jiao Tong University in 2025, Thomas Karam was a postdoctoral researcher at the University of Oxford (2022-2025), a doctoral student at the University of Cambridge (2018-2022), and a student up to Masters level at the Ecole Normale Superieure of Paris (2014-2018). He is interested in several areas related to combinatorics and is currently focusing on the development of the basic theory of the ranks of tensors. His works have been accepted to journals including Journal of the London Mathematical Society, Advances in Combinatorics, Combinatorics Probability Computing, and the SIAM Journal of Discrete Mathematics. He has delivered invited talks in 20 countries, including at several top Chinese and international institutions.

Minimal Cayley graphs with large girth and vanishing independence ratio

Bartłomiej Kielak
Leipzig University

In 1959, Erdős proved the existence of graphs with arbitrarily large girth and chromatic number. Since then, several explicit constructions have been discovered, including the Cayley graphs of Lubotzky, Phillips, and Sarnak (1988).

For minimal Cayley graphs, however, even the existence of examples with chromatic number greater than 4 remained open until 2026, when Davies, Hatzel, and Yepremyan constructed minimal Cayley graphs with arbitrarily large chromatic number. We extend their result by constructing minimal Cayley graphs with arbitrarily large girth and arbitrarily small independence ratio.

Joint work with Igor Balla, Jędrzej Hodor, Krzysztof Kosiorowski, and Daniel Kráľ.

Biography. Bartłomiej Kielak obtained his PhD in 2023 at Jagiellonian University (under the supervision of Bartosz Walczak and Andrzej Grzesik), and has since worked as a postdoctoral researcher in the group of Dan Kráľ, first at Masaryk University in Brno, and currently at Leipzig University. His main field of interest is extremal combinatorics, with emphasis on Turán-type problems. His strongest result would be definitely determinining the uniform Turán density of a complete 4-vertex 3-hypergraph.

Graph Reconstruction: Theory, Efficiency, and Robustness

Hoang La
Université Paris-Saclay

How much information about small parts of a combinatorial structure determines the whole object? We study reconstruction of graphs, uniform hypergraphs, partially ordered sets and words from queries on known subsets of their elements. Each query reveals only partial information about the induced substructure, according to a fixed rule.

I will present reconstruction characterizations for these classes and explain the common form they take. For each class and fixed query size, an explicit finite collection of distinctions between small structures determines which query rules recover every sufficiently large object, with labeled reconstruction or reconstruction up to isomorphism depending on the class. Examples will illustrate both how limited information can suffice and how persistent ambiguities arise when the criteria fail. These results provide a foundation for a common reconstruction theory and raise further questions about efficient recovery and unreliable answers.

Biography. Hoang La is a maître de conférences at Université Paris-Saclay and a member of the GALaC team at LISN. His research concerns structural graph theory, graph reconstruction and partially ordered sets. He obtained his PhD from Université de Montpellier in 2022 and subsequently held a postdoctoral position at Jagiellonian University in Kraków before joining Université Paris-Saclay in 2023.

The Erdős-Gallai bound for consecutive even cycle lengths

Hong Liu
Institute for Basic Science (IBS)

Erdős and Gallai in 1959 proved the seminal result that every n-vertex graph with no cycle of length at least $2t+2$ has at most $\frac{2t+1}{2}(n-1)$ edges. We prove the extension that, for every sufficiently large t, the same quantity is also the sharp extremal bound for graphs with no t consecutive even cycle lengths, resolving a conjecture of Verstraëte. Thus, at the Erdős–Gallai threshold, forcing an entire interval of even cycle lengths costs no more than forcing its longest member. More precisely, every n-vertex graph G with $e(G) \geq \frac{(2t+1)(n-1)}{2}$

- either contains t consecutive even cycle lengths,
- or equality holds and G is connected with every block isomorphic to K_{2t+1}.

As consequences, for every sufficiently large even k we determine the sharp edge thresholds forcing a cycle of length 0 (mod k) or 2 (mod k), answering questions of Bai, Grzesik, Li, and Prorok and of Gao, Li, Ma and Xie, respectively, for sufficiently large even k. The proof develops a stability-enhanced sublinear expander method. Its main new ingredient is a dense-case decomposition that recovers the lengths lost in the expander extraction by combining a flexible dense core with rooted cycle families in the vertices outside the core.

Biography. Hong Liu, the head of the Extremal Combinatorics and Probability Group (ECOPRO) and a distinguished research fellow at the Institute for Basic Science (IBS), he got his Ph.D. in 2015 at University of Illinois at Urbana Champaign, advised by József Balogh. He is interested in extremal and probabilistic combinatorics, graph theory, discrete geometry, Ramsey theory, and combinatorial number theory.

Recent Progress on Hypergraph Ramsey Theory

Ruilong Liu
Shandong University

The Ramsey number $r_k(s,n)$ is the smallest integer N such that every N-vertex k-graph contains either a copy of $K_s^{(k)}$ or an independent set of size n.

We prove that $r_4(5,n) \geq 2^{2^{cn^{1/7}}}$, where $c>0$ is an absolute constant. As a consequence, we determine the tower growth rate of $r_k(k+1,n)$, which completely solves the problem of establishing the tower growth rate for all classical off-diagonal hypergraph Ramsey numbers, first posed by Erdős and Hajnal in 1972.

Biography. Ruilong Liu is a PhD student at Shandong University under the supervision of Prof. Guanghui Wang. His research interests are in Ramsey theory and related areas of combinatorics.

Extensions of the Andrásfai–Erdős–Sós Theorem

Xizhi Liu
University of Science and Technology of China

The classical Andrásfai–Erdős–Sós theorem states that every triangle-free graph with minimum degree greater than $2n/5$ is bipartite. In this talk, I will discuss several sharp extensions of this theorem. These include a strengthening involving both minimum and maximum degrees, as well as hypergraph analogues for generalized triangles. In particular, we establish optimal minimum-degree and positive-codegree conditions that force multipartite structures in hypergraphs. The results are joint work with Sijie Ren and Jian Wang.

Biography. Xizhi Liu is currently a Specially Appointed Professor at the School of Mathematical Sciences, University of Science and Technology of China. Previously, he was a postdoctoral researcher at the University of Warwick, UK. His research focuses on extremal combinatorics and Turán theory for graphs and hypergraphs. His research interests include extremal problems for hypergraphs, stability theory, Lagrangian methods, spectral extremal combinatorics, and probabilistic and structural combinatorics.

Longest cycles in highly connected graphs

Jie Ma
University of Science and Technology of China / Tsinghua University

We will consider several problems on longest cycles in highly connected graph. We establish a higher-connectivity extension of Dirac's theorem and then use it to derive two applications: a conjecture of Bondy and a conjecture of Smith on longest cycles. This is joint with Bo Ning and Ziyuan Zhao.

Biography. Jie Ma's research focuses on extremal and probabilistic combinatorics and graph theory, with applications to theoretical computer science. He earned his B.S. from the University of Science and Technology of China in 2007 and his Ph.D. from Georgia Institute of Technology in 2011 under the supervision of Xingxing Yu. He subsequently held a Hedrick Assistant Professorship at UCLA and a postdoctoral position at Carnegie Mellon University. His honors include the 2020

Hall Medal from the Institute of Combinatorics and its Applications and support from the National Science Fund for Distinguished Young Scholars.

A generalisation of Ramsey's theorem and its applications to χ- and degree-boundedness

Lucas Picasarri-Arrieta
National Institute of Informatics

We prove a new generalisation of Ramsey's theorem by showing that every 2-edge-coloured graph with sufficiently large minimum degree contains a monochromatic induced subgraph whose minimum degree remains large. From this, we also derive that every orientation of a graph with large minimum degree contains either a large transitive tournament or an induced antidirected digraph whose minimum degree is still large. As a consequence, we obtain two general tools showing that certain extensions of degree-bounded graph classes preserve degree-boundedness. A hereditary class $\mathcal{G}$ is degree-bounded if, for every integer s, there exists $d = d(s)$ such that every graph $G \in \mathcal{G}$ either contains $K_{s,s}$ or has minimum degree at most d. With these tools, we obtain for instance that odd-signable graphs and Burling graphs are degree-bounded. We also characterise exactly the oriented graphs F such that the graphs admitting an orientation without any induced copy of F are degree-bounded. This is joint work with Arnab Char and Ken-ichi Kawarabayashi.

Biography. Lucas Picasarri-Arrieta is a postdoctoral researcher at the National Institute of Informatics in Tokyo, where he works with Ken-ichi Kawarabayashi. He received his Ph.D. in Computer Science from Université Côte d'Azur (France) in 2024, under the supervision of Frédéric Havet and Stéphane Bessy. His research interests include graph theory and combinatorics, with a particular focus on digraph colouring, extremal combinatorics, and probabilistic methods. In February 2027, he will join CNRS as a *Chargé de recherche* (Research Fellow) at LAMSADE (Paris).

The PCF Colouring Conjecture is true for large Δ

François Pirot
Université Paris-Saclay

Let G be a graph and c a vertex-colouring of G. We say that a vertex-subset X has a solitary colour x in c if x has a unique occurrence in the multiset $c(X)$. We say that c is conflict-free if every (open) neighbourhood of G has a solitary colour in c, and the pcf-chromatic number of G, denoted $\chi_{pcf}(G)$, is the minimum k such that a proper conflict-free (pcf for short) k-colouring of G exists. In 2023, Caro, Petruševski, and Škrekovski conjectured that $\chi_{pcf}(G) \leq \Delta + 1$ for every connected graph G of maximum degree $\Delta \geq 3$ (PCF Colouring Conjecture). This conjecture has been proven for (almost) regular graphs with sufficiently large Δ by Liu and Reed in 2024 – they require that the minimum degree is at least 0.9999Δ. The goal of this presentation will be to present a simplified version of their proof that works for graphs with minimum degree $\Omega(\log \Delta)$. I will then explain how to make the proof work with no minimum degree condition, with suitable extensions of the arguments to hypergraphs. This resolves the

PCF Colouring Conjecture for graphs of sufficiently large maximum degree. This is joint work with Quentin Chuet, Tianjiao Dai, Hoang La, and Qiancheng Ouyang.

Biography. François Pirot is an associate professor (maître de conférences) at Université Paris-Saclay (France). He received his doctorate in mathematics from Radboud University and in computer science from Université de Lorraine on 4 November 2019. His PhD was supervised by Ross Kang and Jean-Sébastien Sereni, and his thesis received the Charles Delorme Prize, which honours an outstanding thesis in graph theory each year. His main research activities focus on the colouring problem in various contexts, including graph powers, locally sparse graphs, distributed colouring, and randomised algorithms. He has also worked on problems from bioinformatics related to circular codes, using efficient graph-theoretic tools.

Polynomial Bounds for the Graph Minor Structure Theorem

Michał Seweryn
Jagiellonian University

The *Graph Minor Structure Theorem* describes approximately the structure of graphs which do not contain a fixed graph H as a minor; informally, such graphs can be obtained by gluing together smaller pieces, each of which consists of a graph embedded on a surface of small genus augmented with at most $f(t)$ "vortices" of "depth" at most $f(t)$, plus at most $f(t)$ "apices" — where, classically, f was not known to be better than a tower of exponentials. This theorem was proved by Robertson and Seymour as a tool for solving Wagner's Conjecture, and has since found numerous other applications in graph theory, algorithms, and combinatorial optimization.

In my talk, I will give an overview of the theory of graph minors, and discuss a recent proof of the Graph Minor Structure Theorem with polynomial bounds on $f(t)$.

This is joint work with Maximilian Gorsky and Sebastian Wiederrecht.

Biography. Michał Seweryn is an Assistant Professor at Jagiellonian University in Kraków. His research focuses on structural graph theory and partially ordered sets. He did postdocs at ULB with Gwenaël Joret, and at Charles University with Zdeněk Dvořák.

Disjoint cycle problems in digraphs

Yun Wang
Shandong University

In this talk, I will survey the history and recent progress on vertex-disjoint and arc-disjoint cycle problems in digraphs, including vertex-disjoint cycles with length restrictions. I mainly focus on the Bermond–Thomassen Conjecture for vertex-disjoint cycles, the Alon–McDiarmid–Molloy Conjecture for arc-disjoint cycles, and the directed analogue of the El-Zahar Conjecture. I will present some of our contributions to the area and discuss several remaining open problems.

Biography. Yun Wang received her PhD from Shandong University and undertook joint doctoral training at the University of Southern Denmark, under the supervision of Jin Yan and Jørgen

Bang-Jensen. She is currently a postdoctoral researcher at the Institute of Data Science, Shandong University, under the supervision of Guanghui Wang.

Small circumference in regular sublinear expanders

Fan Yang
Institute for Basic Science (IBS)

Sublinear expansion is weak enough to be extracted from arbitrary graphs while retaining nearly all of their average degree, yet it has proved strong enough to force global structures in many sparse extremal problems. Letzter, Methuku and Sudakov [JLMS 2026] proved the existence of nearly Hamilton cycles in sufficiently dense regular sublinear expanders. Montgomery [ICM 2026] subsequently conjectured that, every d-regular sublinear expander with d sufficiently large (but constant) is Hamiltonian. We disprove this conjecture in a strong form by constructing n-vertex d-regular sublinear expanders with $d = \left(\frac{1}{2} + o(1)\right) \log^2 n$, which can forbid any cycle covering an arbitrarily small given positive constant portion of its vertices. The construction blows up one side of a biregular Ramanujan graph into almost-complete blocks and keeps the other side as a sparse vertex separator.

We also prove a similar statement for a closely related notion of edge expanders. For every sufficiently small $\gamma > 0$, there is an infinite family of n-vertex d-regular γ-edge-expanders with $d = \Theta(\gamma^{-1})$ and circumference $O(\gamma n)$, matching the standard lower bound $\Omega(\gamma n)$. The construction also comes from an expanding regular core such that each vertex has an almost-complete graph attached to it. The core guarantees edge expansion, while the single-vertex attachments confine every cycle.

Biography. Fan Yang is a postdoctoral fellow at the Institute for Basic Science (IBS), working with Professor Hong Liu with support from the China Scholarship Council (CSC). She received her Ph.D. from Shandong University in 2022 and has published nine research papers.